Spatio-Temporal Filtering Approach for Tomographic SAR Data

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***Abstract*— Synthetic aperture radar tomography (TomoSAR) has recently received particular interest from the remote-sensing community, due to its ability to provide 3-D reconstructions of environments with complex structures. Unfortunately, different forms of decorrelations and processing errors affect the quality of the resulting 2-D/3-D images. One way to cope with the impact of these nuisances is to apply appropriate filtering to the interferometric data stack as a preprocessing step. The first obstacle to be dealt with, especially in urban areas, is to define a filter whose parameters have to be set in such a way as to improve smoothing capabilities while preserving edges. To this aim, the main objective of this article is twofold: 1) the application of a spatio-temporal contextual filter whose parameters depend on 3-D quality indicators of the multibaseline interferometric image stack and 2) evaluation of the denoising effect on the application of nonparametric spectral estimation and detection algorithms. For that, we consider several quantitative metrics to assess, on the one hand, the filtering performances, and, on the other hand, its impact on the reflectivity function recovered from conventional tomographic inversion and detection methods. Experimental results from a set of TerraSAR-X (TSX) images highlight the efficiency of the filtering process by improving the scatterers' detection and height localization, of a man-made structure.**

***Index Terms*— 3-D quality maps, building reconstruction, height cloud map, spatio-temporal filtering, synthetic aperture radar tomography (TomoSAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) systems provide an efficient way for collecting information on the physical and geometrical characteristics of objects [1]. Their emergence in the field of remote sensing has led to a breakthrough in a multitude of applications, in particular, in 3-D, 4-D, and 5-D surface mapping and monitoring [2], [3], [4].

SAR tomography (TomoSAR) has been extensively developed in the last two decades as a 3-D SAR imaging technique and allowed the synthetic aperture principle to be extended to the third dimension, taking into account the elevation information along the axis perpendicular to the azimuth-range plane [2], [5]. As a result, the retrieval of 3-D reflectivity profiles from a set of images acquired with an interferometric multibaseline configuration [6] has become possible with good precision [7].

Among the objectives of TomoSAR is to solve the layover phenomenon [8], particularly in the case of areas characterized by a high density of vertical structures (buildings, towers, skyscrapers, etc.). To solve this problem, both the amplitude and phase of the single-look complex (SLC) images need to be exploited. The use of advanced signal-processing techniques with enhanced resolution capabilities and sidelobes suppression enables solving the inversion problem with increased accuracy in the elevation direction [9]. It is possible to categorize TomoSAR processing methods into three types: parametric–nonparametric spectral estimation and compressive sensing (CS) approaches.

The most popular parametric focusing methods are Minimum-Norm [8], Non-Linear Least-Square [10], [11], M-Relax [12], and MUSIC [8], [13]. They are based on a predefined model [14] and offer good results when the latter fits well with the observed data [12]. Nonparametric approaches [5], such as Beamforming [8], Singular Value Decomposition [7], [15], CAPON [6], and Linear Prediction [16], instead, do not require any a priori information on targets and provide a fast and robust estimation, especially when a large number of images is available [10], [17]. As for CS-based techniques, they exploit the sparsity property of SAR signals and can achieve vertical resolution beyond the Rayleigh limit [18], [19], [20]. Due to their complexity, the development of classical methods remains a desirable solution.

An additional important issue to be considered in SAR tomographic processing is the reduction of the disturbances and distortions affecting the multibaseline interferometric images and degrading the reflectivity reconstruction. These disturbances are the outcome of:

1) the geometry of the acquisition configuration, making areas where the phase is completely random unusable (i.e., shadowed areas);
2) the effect of decorrelations, which can be classified into: a) *spatial decorrelation*, due mainly to the elementary reflection variability caused by the view angle changes from one pass to another; b) *temporal decorrelation*, due to changes in atmospheric conditions and the physical and geometric properties of the scatterers during the

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time interval between the different acquisitions; and c) *thermal decorrelation*, due to additive thermal noise generated in the electronic devices of the acquisition system.

These disturbances can lead to false scatterers' identification. Pauciullo et al. [21] proposed a detector, based on a sequential generalized likelihood ratio test (GLRT), capable of estimating the number of scatterers present in each resolution cell and their respective elevations if any. Afterward, different variants of GLRT have been proposed [22], [23], [24] to retrieve the scatterers' heights and velocities (3-D and 4-D maps) [25]. Another way can be considered to deal with the noise effect by reducing it in the preprocessing step using a properly designed filter. The main obstacle encountered in the filter design is finding a filtering algorithm that adapts to the complexity of SAR data while meeting the expected accuracy requirements on the results.

A multitude of filtering algorithms have been proposed in the literature to reduce the effect of noise on the interferometric SAR (InSAR) phase for single-polarization and multipolarization SAR (PolSAR) data [26], [27]. One of the most widely used InSAR phase filters is the Goldstein filter [28], [29], due to its remarkable noise suppression capability and fast operation. In TomoSAR framework, D'Hondt et al. [30] studied the efficiency of nonlocal means (NLM) filters, in estimating the multidimensional covariance matrix (MCM) by considering an adequate nonlocal neighborhood. Shi et al. [31] performed two stages of multibaseline data filtering. In the first stage, a classic principal component analysis (PCA) approach was used. It consists in projecting the SAR images on a new space as a means to facilitate the separation of useful information from noise. The second stage consisted of applying the NLM filter for neighborhood redefinition. None of the above-mentioned studies had considered filtering the SAR data as a preprocessing step for the purpose of tomographic inversion.

In this article, our work focuses on improving the rendering of 3-D structures reconstruction from tomographic data by reducing the noise influence. To do this, the first step of our study involves a new adaptive implementation of the Goldstein filter [28], where its alpha parameter is estimated from 3-D quality maps. The filtering process is applied to complex interferograms rearranged in accordance with a spatio-temporal context, which gives the potential of providing a better outcome by including the temporal dimension. The effectiveness of such a denoising method in the context of TomoSAR has not been studied before. In the second step, we evaluate the impact of the proposed contextual spatio-temporal filtering on urban tomographic reconstruction processing with a low number of acquisitions. So, accordingly, both spectral estimation and GLRT-based detection methods are implemented to highlight the potential of the denoising process.

The remainder of the article is structured as follows: Section II presents a brief overview of TomoSAR principal and techniques. Section III describes our proposed contextual spatio-temporal filtering algorithm by highlighting our contributions namely the 3-D neighborhood definition and restructuring, as well as the 3-D quality maps exploit. Results on real TerraSAR-X (TSX) data are presented in Section IV, where the influence of the denoising approach on TomoSAR detection along with reflectivity profile reconstruction are evaluated qualitatively and quantitatively. Finally, Section V concludes this article.

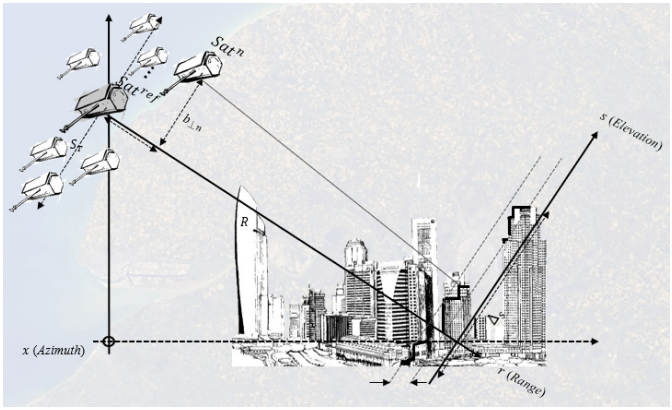


Fig. 1. Illustration of TomoSAR geometry.

II. TomoSAR Theory

A. Tomographic Data Model

Due to the complexity of SAR radar systems, the raw data acquired by the receiving platform is a hologram [32], and therefore a focusing step is required to form a 2-D azimuth-range SAR image. In this regard, two hypotheses are assumed: the first supposes that the scatterers are temporarily stationary and remain in the far field of the sensor antenna, and the second considers that the imaged scene is a set of scattering points whose backscattered responses are superimposed linearly. Under the latter hypothesis, known by Born approximation [7], the linear operator characterizing the SAR imaging process is the geometric projection of the 3-D objects on the 2-D space (the azimuth x, the range r) [1]. This projection is the reason why the information on the spatial structure and the scatterers' location in the same resolution cell are confused [5]. Thus, a conventional SAR image provides a single 2-D tomographic slice of these objects.

Consequently, to perform a 3-D reconstruction of each azimuth-range pixel, an additional synthetic aperture along the elevation direction s, perpendicular to the (x, r) plane, is carried out by acquiring the signal along N trajectories (see Fig. 1).

In a fixed azimuth-range pixel, the nth image ($n = 1, \ldots, N$) acquired with a perpendicular baseline $b_{\perp n}$, can be expressed as the coherent superposition of the signals backscattered from all the scatterers lying in the considered resolution cell. This superposition value g_n can be written as [7]

$$g(\xi_n) = \int_{-\frac{\Delta_s}{2}}^{+\frac{\Delta_s}{2}} \gamma(s) \exp(-j2\pi\xi_n s) ds + w_n \tag{1}$$

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with a spatial frequency $\xi_n = -2b_{\perp n}/\lambda R$, where λ is the operating wavelength, Δ_s is the elevation extension of the scene, and w_n models the additive noise. This disturbance can be described by a complex circularly Gaussian random variable, with zero-mean and variance σ_w^2.

From (1), it can be observed that $g(\xi_n)$ is a sample of the Fourier transform of the reflectivity profile $\gamma(s)$ along s. In discretized matrix form, obtained by uniformly sampling the elevation in M values, (1) can be written as

$$\boldsymbol{g} = \mathbf{A}\boldsymbol{\gamma} + \boldsymbol{w} \tag{2}$$

where $\boldsymbol{g}$ is the $N \times 1$ multibaseline data vector, $\boldsymbol{\gamma}$ is the $M \times 1$ reflectivity vector containing the reflectivity samples in the discretized elevation values s_m, $\boldsymbol{w}$ is the noise vector, and $\mathbf{A}$ is the $N \times M$ sensing or steering matrix, whose mth column is the steering vector

$$\boldsymbol{a}(s_m) = \left[e^{-j2\pi\xi_1 s_m}\ e^{-j2\pi\xi_2 s_m}\ \cdots\ e^{-j2\pi\xi_N s_m}\right]^T \tag{3}$$

with ^{T} denoting the transpose and $m = 1, \ldots, M$.

Therefore, the recovery of the elevation profile $\boldsymbol{\gamma}$ of the scattering targets in each azimuth-range pixel involves an inversion of the linear equation system (2). Depending on the considered scenario [9], which, in our case, consists of man-made structures in urban areas, a discrete reflectivity profile can be described by a deterministic surface scattering model.

B. TomoSAR Techniques

Tomographic inversion can be applied on one azimuthal (or range) line to generate 2-D tomograms, or alternatively on a set of pixels meeting certain conditions, such as permanent scatterers (PSs) condition, to generate reflectivity and height maps only for the scatterers remaining sufficiently coherent (to avoid noisy pixel). A second tomographic process includes a GLRT-based detector aiming to discriminate between different hypotheses.

1) Spectral Estimators: Spectral estimators are widely used in TomoSAR to recover the power distribution profile $P(s_m) = |\gamma(s_m)|^2$, which involves the analysis of the covariance matrix $\mathbf{R_g}$.

Robust-CAPON beamforming [33], [34] is a nonparametric method that offers good performance regarding elevation resolution and sidelobes removal. Basically, it consists of scanning the elevation interval while measuring the power received from the backscattered signals, and taking advantage of the diagonal loading of the MCM to mitigate the self-cancellation phenomenon [17]. The reconstructed profile formula is

$$\hat{P}_{\text{CAPON}}(s_m) = \left|(\boldsymbol{a}(s_m)^H \cdot \mathbf{R}_g^{-1} \cdot \boldsymbol{a}(s_m))^{-1}\right|^2. \tag{4}$$

Parametric MUSIC [12], [13], [33] estimator requires the number of scatterers, to decompose the estimated MCM into subspaces. Under the assumption that the noise space $\mathbf{E}$ is orthogonal to the data of interest [9]. This method ensures good resolution in the case of uncorrelated signals. The backscattering profile expression is

$$\hat{P}_{\text{MUSIC}}(s_m) = \left|(\boldsymbol{a}(s_m)^H \cdot \mathbf{E}^H \cdot \mathbf{E} \cdot \boldsymbol{a}(s_m))^{-1}\right|^2. \tag{5}$$

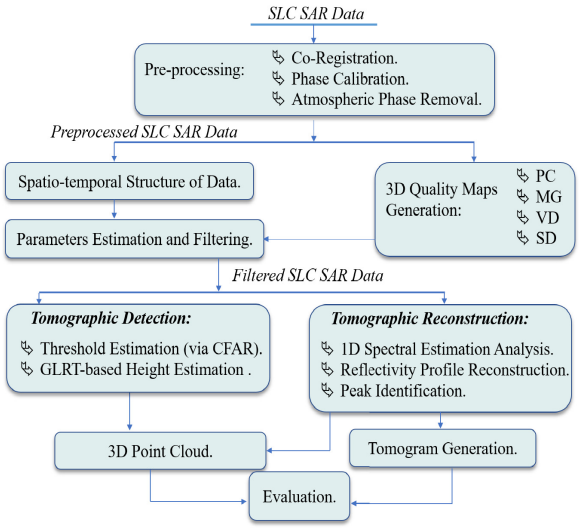


Fig. 2. Block diagram of the proposed method.

2) GLRT-Based Detectors: Assuming elevation profiles are sparse in urban areas [19], and searching for the dominant coherent scatterer in each resolution cell, the detection problem in accordance with (2) can be formulated in two statistical hypotheses: (H_0) for the absence of a point scatterer and (H_1) for the presence of a point scatterer

$$\boldsymbol{g} = \begin{cases} \boldsymbol{w}, & H_0 \\ \boldsymbol{a}(s_m)\boldsymbol{\gamma} + \boldsymbol{w}, & H_1. \end{cases} \tag{6}$$

The hypotheses can be determined by applying the following GLRT test [22], [35]:

$$\Lambda = \frac{\max\limits_{s_m} \frac{1}{N}|\boldsymbol{a}(s_m)^H \boldsymbol{g}|^2}{\boldsymbol{g}^H \boldsymbol{g}} \underset{H_0}{\overset{H_1}{\gtrless}} T_s \tag{7}$$

where T_s is the threshold derived using a constant false alarm rate approach. When the number of observations is too small, the detector performance may be unsatisfactory, thus introducing L look GLRT can improve the outcome, under the presumption that the selected pixel neighbors exhibit similar characteristics. In this case, the multilook GLRT test becomes [24], [34]

$$\Lambda_{\text{ML}} = \frac{\max\limits_{s_m} \sum\limits_{l=1}^{L} \frac{1}{N}|\boldsymbol{a}(s_m)^H \boldsymbol{g}_l|^2}{\sum\limits_{l=1}^{L} \boldsymbol{g}_l^H \boldsymbol{g}_l} \underset{H_0}{\overset{H_1}{\gtrless}} T_{\text{ML}}. \tag{8}$$

III. Proposed Approach

The proposed tomographic processing is schematized by the flowchart in Fig. 2. In this section, we describe the main blocks of this process and highlight our contributions.

A. Preprocessing of SAR Data

The processing of tomographic data poses a number of challenges compared to the various conventional treatments.

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Obtaining good vertical resolution with the absence of sidelobes is often a difficult task to accomplish, due to the acquisition system constraints, hence the need to use more advanced 1-D, 2-D, and 3-D signal-processing techniques.

First, a geometric co-registration [36], [37] in three dimensions is performed according to a reference master image to ensure that each target contributes to the same pixel in all the SAR images at a subpixel scale. In the second place, a phase calibration is necessary to recompensate for the disturbances resulting from space-varying phase [38], on the one hand, and atmospheric delay [15], [39], on the other hand.

B. Proposed Filtering Process

The design of a tomographic filter has to be made in such a way as to satisfy the following requirements.

1) *Preservation of edges:* In urban environments, building and artificial structures have distinct backscattering profiles with considerably variable intensities, and therefore, it is essential to avoid mixing the scattering information of separate physical adjacent objects.
2) *Preservation of the interferometric phase:* TomoSAR exploits the signal phase, which is related to the geometric location of the scatterers. Then, the interferometric phase must be preserved, particularly in urban areas where vertical structures are characterized by a significant spatial variation of the phase.
3) *Reduction of decorrelations' effect:* It is important that the filter adapts enough to be able to reduce the noise effects from different types of decorrelations.
4) *Maximum smoothing of flat areas:* After removing the flat earth phase, the flat areas will be characterized by nearly constant phase and reflectivity, and therefore they must be well smoothed.

The proposed filtering process is based on three steps.

1) Spatio-Temporal Structure of Data Space: During our application, multitemporal multibaseline SAR data is adaptively structured when defining the filter window. The latter is segmented into overlapping patches with an interlaced resolution cell arrangement to exploit the 2-D/3-D interpixel and intrapixel relationship. The exploitation of interpixel correlation makes it possible to reduce geometric geolocation and registration errors, while the exploitation of the 2-D/3-D intrapixel relationship reduces the effect of multilooking on the spatial resolution, by emphasizing the backscattering properties from adjacent targets in 3-D space. An illustration of this organization is shown schematically in Fig. 3.

2) 3-D Quality Map Generation: The 3-D quality maps were proposed in the context of SAR imagery initially for the phase unwrapping [40]. They are calculated using image-processing techniques. Their role is to label each pixel in the image by its quality in a well-defined neighborhood context. In our case, we are interested in the quality of the edges in the amplitude (Am^n, the absolute value) and phase (φ^n, the interferometric phase value) images. Therefore, the most effective indicators according to the literature are: the pseudo-correlation (PC), the maximum of the gradient, the variance of the derivative (VD), and the second derivative (SD).

PC quality indicator measures the degree of local correlation between the phases of adjacent pixels. It highlights regions with sudden phase variation, where these regions will have low values in the quality map, and its expression is

$$\mathrm{PC}^n(x,r) = \frac{1}{C_w}\sqrt{\left(\sum_i\sum_j\sum_k \cos(\varphi_{i,j,k})\right)^2 + \left(\sum_i\sum_j\sum_k \sin(\varphi_{i,j,k})\right)^2} \tag{9}$$

where C_w is related to the considered neighborhood context and equals the cube of the moving window extent. The letters i, j, and k are the indices according to the azimuth, range, and temporal neighborhood of the considered window C_w.

The maximum gradient (MG) quality indicator measures the amplitude of the largest gradient corresponding to the maximum of all partial derivatives of the surrounding pixels in a specific defined neighborhood. It provides a distinction between regions with weak phase variation and transition areas, and the calculation formula for which is

$$\mathrm{MG}^n(x,r) = \max\begin{cases}\max\{|\nabla^x(i,j,k)|\}_{C_w}\\ \max\{|\nabla^r(i,j,k)|\}_{C_w}\\ \max\{|\nabla^n(i,j,k)|\}_{C_w}.\end{cases} \tag{10}$$

The terms ∇^x, ∇^r, and ∇^n are the partial derivatives of the complex pixel according to the azimuth, range, and temporal direction, respectively.

The VD quality indicator measures the local statistical variance of the pixel's derivatives. Its expression is

$$\mathrm{VD}^n(x,r) = \frac{1}{C_w}\left(\begin{array}{l}\left(\sqrt{\sum_i\sum_j\sum_k(\nabla^x(i,j,k)-\bar{\nabla}^x)^2}\right.\\ +\sqrt{\sum_i\sum_j\sum_k(\nabla^r(i,j,k)-\bar{\nabla}^r)^2}\\ \left.+\sqrt{\sum_i\sum_j\sum_k(\nabla^n(i,j,k)-\bar{\nabla}^n)^2}\right)\end{array}\right). \tag{11}$$

The terms $\bar{\nabla}^x$, $\bar{\nabla}^r$, $\bar{\nabla}^n$ are the partial derivatives means.

The SD quality indicator measures the second difference of each complex pixel in the azimuth, range, temporal, and diagonal directions. It provides information about abrupt transitions of a pixel inside a 3-D window. The calculation formula is

$$\mathrm{SD}^n(x,r) = \sqrt{\mathrm{az}^n(x,r)^2 + \mathrm{rg}^n(x,r)^2 + \mathrm{nr}^n(x,r)^2 + \sum_i D_i^n(x,r)^2} \tag{12}$$

where az, rg, nr, and D are the second difference images according to the azimuth, range, temporal, and diagonal direction, respectively. Their corresponding expressions can be found in the appendix.

3) Filter Parameters Estimation and Design Definition: The Goldstein filter [28] and its improvements [29], [41] [42] are widely used in InSAR imaging because of their efficiency and robustness. The nonlinearity of those filters is also a big asset through their integration of a contextual weighting factor linked to the original data. About a hundred papers have been

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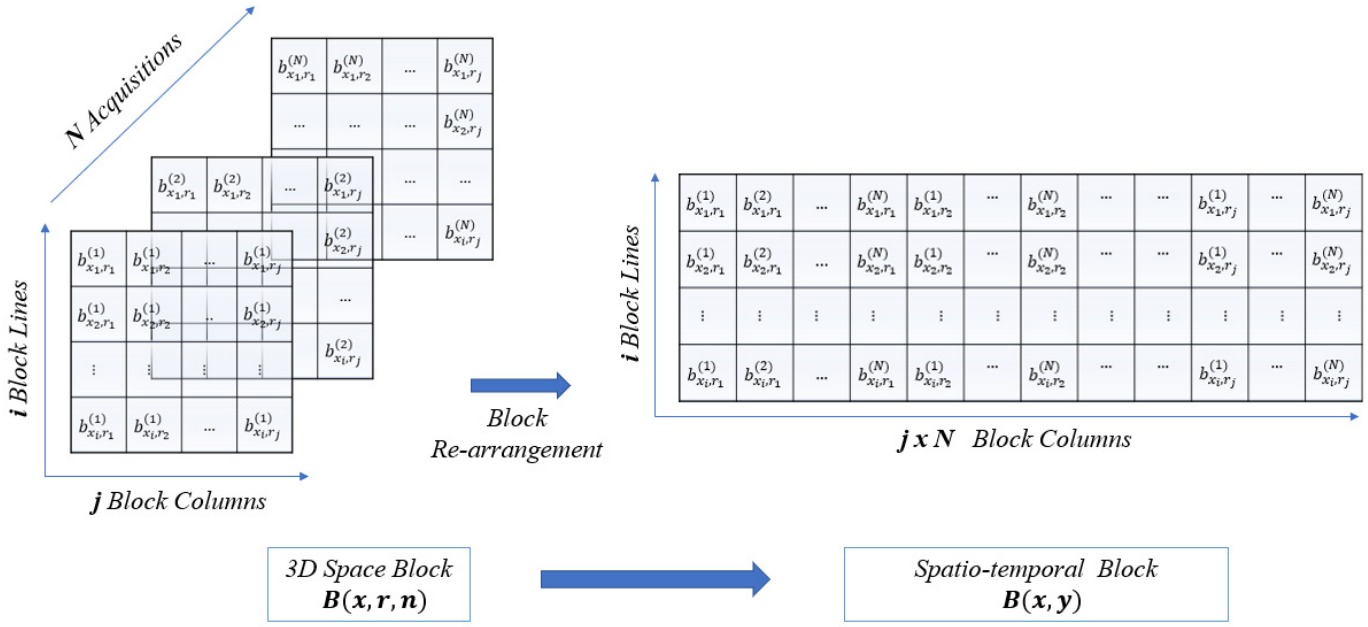


Fig. 3. Spatio-temporal structure for block rearrangement (the notation $b_{x_i,r_j}^{(N)}$ stands for a pixel from the Nth acquisition, at azimuth position i, and range position j).

published that focus on estimating the alpha parameter that allows the best fit to different SAR applications and proposing improvements to this estimate. Goldstein filter and its variants have been applied on interferograms and differential interferograms [43], [44].

Based on the same principle and exploring both the amplitude and the phase of the complex interferogram, we generate the local power spectrum $\mathrm{SP}(u, v)$ by calculating the FFT for each complex-valued spatio-temporal block $B(x, y)$ organized according to the proposed context (see Fig. 3): $\mathrm{SP}(u, v) = \mathrm{FFT}\{B(x, y)\}$. Subsequently, we reduce the variance of this spectral estimate by low-pass filtering (in our case, a Gaussian filter whose kernel h has a size of 5×5): $\widetilde{\mathrm{SP}}(u, v) = |\mathrm{SP}(u, v)| * h(u, v)$ ($*$ designates the frequency convolution operation). The constructed nonlinear adaptive contextual filter is given by $\mathrm{SP}(u, v) \cdot \mathrm{SF}(u, v)$, such as $\mathrm{SF}(u, v) = [\widetilde{\mathrm{SP}}(u, v)]^{\alpha}$, with α a parameter to be properly set.

Finally, an inverse Fourier transform is applied to find the values of the filtered pixels

$$\tilde{B}(x, y) = \mathrm{FFT}^{-1}\{\mathrm{SP}(u, v) \cdot \mathrm{SF}(u, v)\}. \tag{13}$$

In this design, our contribution is essentially a new estimate of the parameter α depending on the quality indicators from the 3-D quality maps (PC, MG, VD, and SD) arranged according to the same spatio-temporal neighborhood context of the block to be filtered. Four different filters can be applied by setting the parameter α on the basis of the four different quality maps. From the PC and SD quality maps, the value of α is set equal to the value of the PC module and to the value of the SD. Distinctively, for MG and VD quality indicators, the value of α is set equal to their inverses. Small quality indicators describe regions with abrupt and accentuated transitions where the edges and structures outlines are found. In this case, α is closer to zero and thus the SF coefficients became closer to 1 which means almost no filtering is applied. On the contrary, regions with good quality indicators provide strong values of α, which leads to the maximum amount of smoothing in those areas.

Moreover, for a better outcome, the block size needs to be determined with precaution, since the larger the block size, the less biased the estimation of the mean quality indicators. On the other hand, the amplitude and the phase variation should not be significant between the first and the last pixels in the considered block. Also, having a power of 2 as size is so desirable for the FFT calculation. Therefore, in order to meet those requirements, the size of 8×8 is defined as a good tradeoff with overlapping of 50% between adjacent blocks.

C. Performance Analysis

To quantitatively evaluate the effectiveness of the filtering approach for both compliance with tomographic filter requirements and its impact on scatterers detection in 3-D reconstructions from real data, we calculate six metrics: edge preservation index (EPI), mean square relative error (MSRE), standard deviation (Std), completeness (C_t), *R-squared* (R_q^2), and detection rate. Note that the PS points are identified using the correlation threshold approach, where the temporal dispersion index D_s is calculated. The pixels exhibiting D_s values smaller than 0.25 are eventually the ones that get selected during PS detection [45].

EPI is a relative measure that provides an assessment of the edges' preservation capability on amplitude and phase, respectively. It is calculated after applying a high-pass filter (generally using Laplacian kernel) to generate the edge image of the amplitude and phase component. The EPI formula [46] is (14) and (15), as shown at the bottom of the next page, where AE_f, PE_f, AE, and PE are the amplitude and the phase

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TABLE I
TSX SYSTEM PARAMETERS OF OUR DATASET

Parameters	Quantity
site	Barcelona
Period	$2009 \sim 2010$
Antenna Looking Direction	Right
Pass direction	Ascending
wavelength	0.031 [m]
View angle	35°
Range Distance	618 [km]
Baseline Spam	157.74 [m]
Rayleigh Elevation Resolution	≈ 60 [m]
Height resolution	≈ 30 [m]
Number of images	9

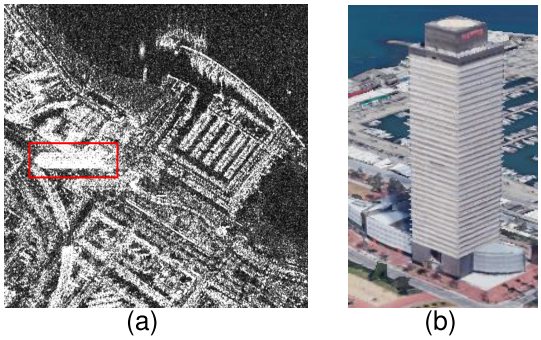


Fig. 4. (a) Intensity SAR image of the studied area (the horizontal and vertical axes represent the range and azimuth, respectively). (b) Optical 3-D image of the tower in Google Earth.

TABLE II
FILTERING EVALUATION PARAMETERS

Parameters	EPI		MSRE	Std	
	EPI_a	EPI_φ		σ_{Am}	σ_φ
Non-filtered	/	/	/	0.9497	0.4409
IG filter	0.2945	0.8200	0.0314	0.9475	0.3324
PCA filter	0.9014	0.6147	0.0298	0.9923	0.3777
PC based filter	**0.9209**	0.7111	0.0365	0.9272	0.3497
MG based filter	0.1411	0.7365	0.0299	0.2075	0.3105
VD based filter	0.1546	0.7128	0.0282	0.2604	0.3039
SD based filter	0.1910	**0.8903**	**0.0277**	**0.1838**	**0.3038**

of the filtered and the original edge image, respectively. The closer the value to 1, the more the edge is preserved in the filtered images.

MSRE [47], [48] measures the average distance between the filtered phases φ_f and the original ones φ, and its expression is

$$\text{MSRE}_\varphi^n = \frac{1}{\dim} \sum \exp\left(j\varphi_f^n(x,r) - j\varphi^n(x,r)\right)^2 \quad (16)$$

where dim is the image size.

TABLE III
DETECTION RATE

Filter	PS Percentage (%)	GLRT Percentage (%)
No filter	17.78	10.80
PCA filter	32.84	08.42
IG filter	20.74	14.02
PC based filter	19.19	13.74
MG based filter	18.93	15.63
VD based filter	18.91	15.64
SD based filter	20.55	15.84

TABLE IV
COMPLETENESS AND *R-Squared* PARAMETERS

Methods	Completeness C_t		R-*squared* R_q^2	
	Non-filtered	filtered	Non-filtered	filtered
Robust-CAPON	**0.2852**	**0.1796**	**0.5890**	**0.6174**
MUSIC	0.4044	0.3177	0.3267	0.3490

The Std matrix represents the degree of amplitude Am and phase φ smoothing. Maximum smoothing is reached when the value of Std tends toward zero, it is calculated as

$$\sigma_{\text{Am}}^n = \frac{1}{\dim}\sqrt{\sum_{i,j}\left(\text{Am}^n(i,j) - \overline{\text{Am}^n}(i,j)\right)^2} \quad (17)$$

$$\sigma_\varphi^n = \frac{1}{\dim}\sqrt{\sum_{i,j}\left(\varphi^n(i,j) - \overline{\varphi^n}(i,j)\right)^2} \quad (18)$$

where $\overline{\text{Am}}$ and $\overline{\varphi}$ are the average values of the filtered amplitudes and phases according to a moving window.

Completeness [30] is a measure of how real scatterers are modeled by estimated ones, and its expression is

$$C_t = \frac{1}{M}\sum_{i=1}^{M} \min_k \quad \text{dist}(\hat{\gamma}(s_k), \gamma(s_i)) \quad (19)$$

where γ and $\hat{\gamma}$ correspond to the ground truth and the estimated reflectivity values, respectively, while dist represents the distortion measure calculated using the Manhattan distance.

R-squared [49] measures how well the estimated heights approximate the actual ones. Its values lie between 0 and 1, and the higher the value, the better the estimation. The calculation formula is

$$R_q^2 = 1 - \frac{\sum_i (s_i - \hat{s}_i)^2}{\sum_i (s_i - \bar{s}_i)^2} \quad (20)$$

where s_i, $\hat{s}_i$, and $\bar{s}_i$ account for the ground truth as well as the estimated and the mean elevation values, respectively.

IV. RESULTS AND EVALUATION

A. Dataset

The proposed approach was applied to a set of SLC images acquired by the TSX satellite sensor. The study area belongs to a high-elevation building having large facades, named Mapfre

$$\text{EPI}_a^n = \frac{\sum\left|\text{AE}_f^n(x,r) - \text{AE}_f^n(x+1,r)\right| + \left|\text{AE}_f^n(x,r) - \text{AE}_f^n(x,r+1)\right|}{\sum|\text{AE}^n(x,r) - \text{AE}^n(x+1,r)| + |\text{AE}^n(x,r) - \text{AE}^n(x,r+1)|} \quad (14)$$

$$\text{EPI}_\varphi^n = \frac{\sum\left|\text{PE}_f^n(x,r) - \text{PE}_f^n(x+1,r)\right| + \left|\text{PE}_f^n(x,r) - \text{PE}_f^n(x,r+1)\right|}{\sum|\text{PE}^n(x,r) - \text{PE}^n(x+1,r)| + |\text{PE}^n(x,r) - \text{PE}^n(x,r+1)|} \quad (15)$$

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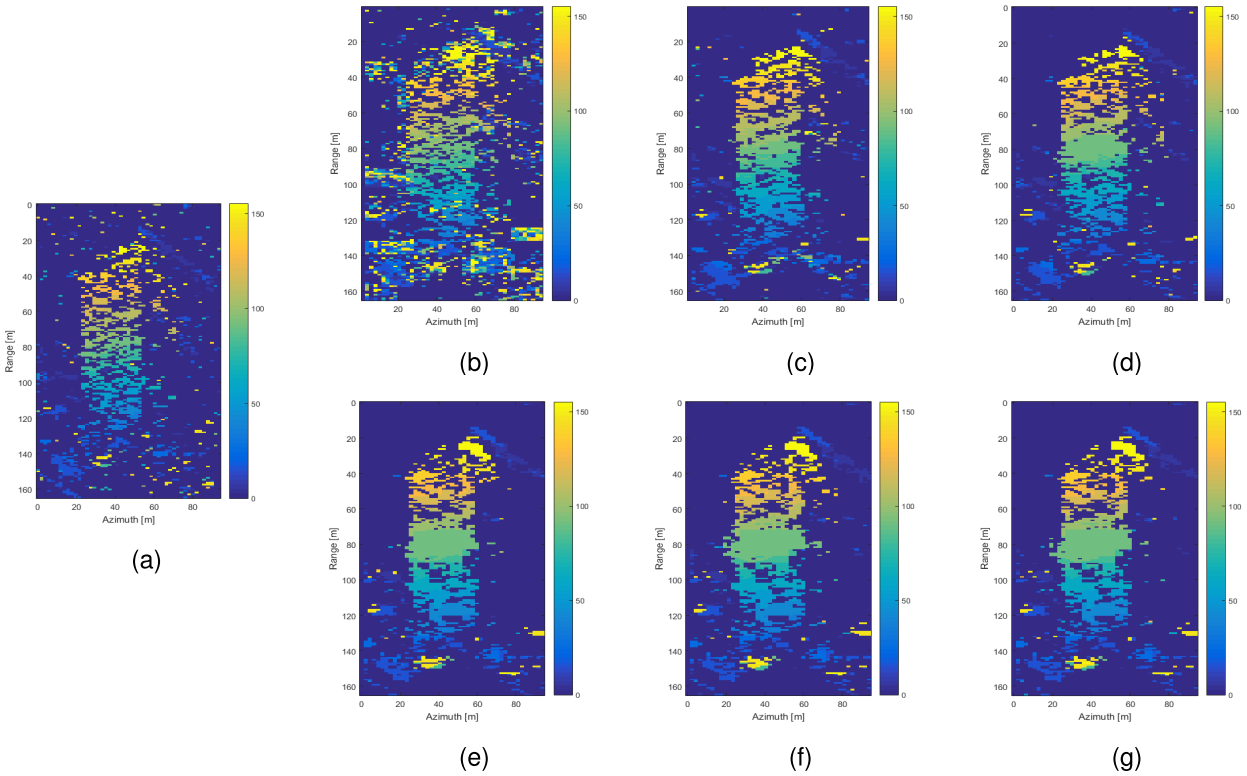


Fig. 5. PS point cloud map applied on (a) nonfiltered data, (b) PCA-based filtered data, (c) IG-based filtered data, (d) PC-based filtered data, (e) MG-based filtered data, (f) VD-based filtered data, and (g) SD-based filtered data.

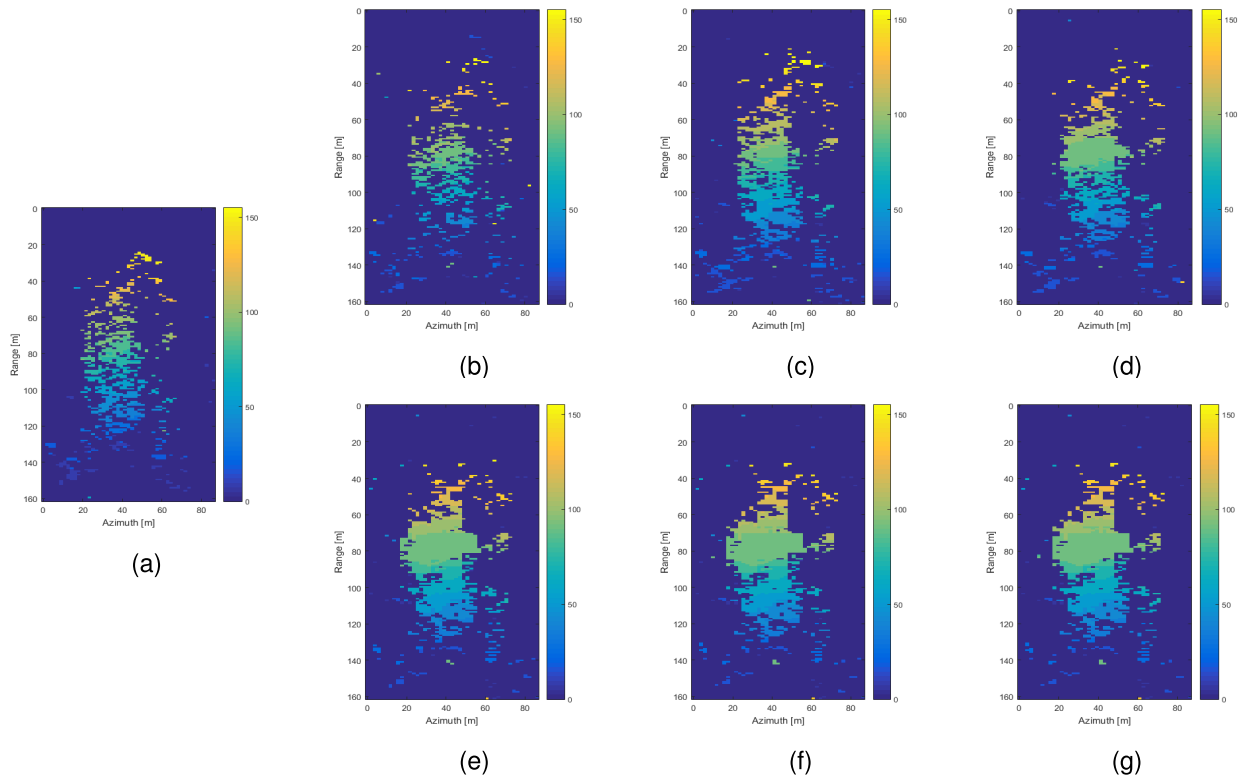


Fig. 6. Height map generated from GLRT-based detector applied on (a) nonfiltered data, (b) PCA-based filtered data, (c) IG-based filtered data, (d) PC-based filtered data, (e) MG-based filtered data, (f) VD-based filtered data, and (g) SD-based filtered data.

Tower, located at a latitude of 41° 23' 16" N and a longitude of 2° 11' 51" E. The system parameters are illustrated in Table I. The intensity map and the location of the tower on Google Earth are illustrated in Fig. 4.

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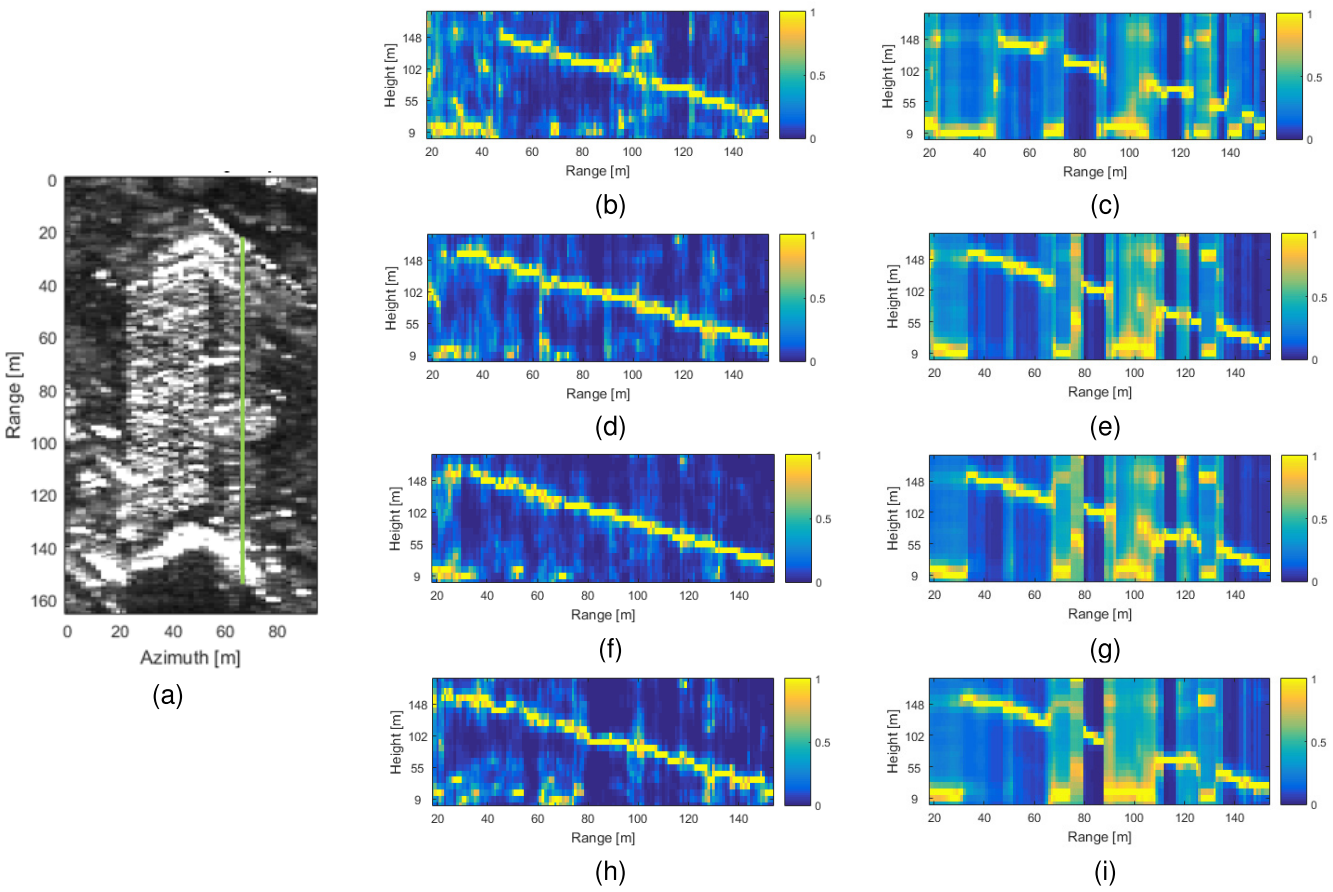


Fig. 7. (a) Mean amplitude map with the considered profile. (b)–(i) Implementation of Robust-CAPON and MUSIC spectral estimators, respectively, applied on (b) and (c) nonfiltered data, (d) and (e) PCA-based filtered data, (f) and (g) IG-based filtered data, and (h) and (i) SD-based filtered data.

B. Evaluation Analysis

Our analysis is divided into three parts: the first is to compare the four contextual nonlinear spatio-temporal filters proposed in Section III-B, along with the conventional filters: improved Goldstein (IG) and PCA. The goal is to determine to what extent the tomographic filtering requirements have been met, mainly referring to the preservation of the edges in amplitude and phase. For this, we used the metrics EPI, Std, and MSRE. The calculation was carried out on nonfiltered and filtered preprocessed images.

The second part shows the contribution of spatio-temporal filtering to tomographic reconstruction and detection. Therefore, we applied the tomographic inversion on all the filtered and nonfiltered images and then we generated a height point cloud. We compared the results obtained qualitatively and quantitatively to select the best filter among the four proposed.

In the last phase, we examine the impact of tomographic denoising further, on the reconstruction of the reflectivity function, through the calculation of the metrics C_t and R_q^2, as well as the generation of tomograms along with 3-D point clouds for scene interpretation and analysis.

1) Performance of the Four Variants of Spatio-Temporal Filtering: The quality of the conventional and designed filters can be expressed and verified by the introduced metrics. They describe the amount of information that get eliminated and the reliability degree of the information retained by the filtering process. The EPI and MSRE parameters quantify the contours restoration in amplitude and in phase, respectively, while Std describes the amount of smoothing. From the calculation results shown in Table II, it is observed that the highest value of EPI_a corresponds to the filter based on the PC quality indicator applied on the edges' images of the amplitude component; on the other hand, the highest value of EPI_φ and the lowest value of MSRE correspond to the filter based on the SD quality indicator applied on the edges' images of the phase component, and this attests its capacity to preserve the phase outlines.

The results of σ_{Am} and σ_φ show that the SD-based filter provides the best values compared to the ones obtained by the original nonfiltered data which acclaim that this filter offers the maximum smoothing in flatted areas. We can also perceive that the filters based on VD and MG quality indicators furnish good values of σ_{Am}; however, despite having the best edges' preservation of the amplitude according to EPI_a, the PC-based filter shows a poor smoothing capability (high value of σ_{Am}), which suggest that no distinction between the outlines structures and the flatted zones is made. From those metrics analyses, we can conclude that the closest filter design to the requirements of TomoSAR filtering is the one based on the SD quality indicator. In addition, the MG- and the VD-based filters can offer a good alternative. As for the conventional denoising methods, it is observed that the IG filter performs well on the phase component according to EPI_φ, σ_φ and MSRE, compared

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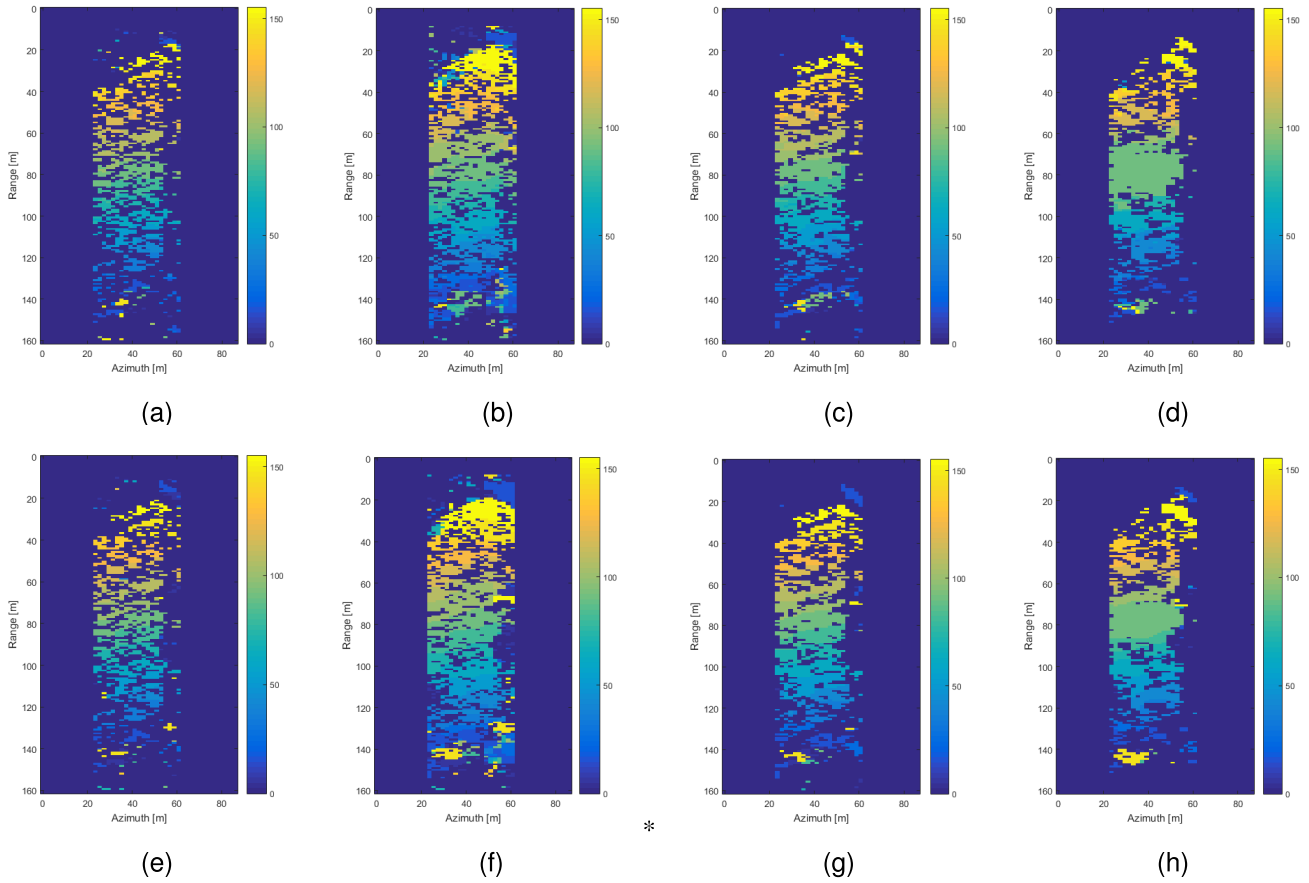


Fig. 8. PS point cloud map generated using (a)–(d) Robust-CAPON and (e)–(h) MUSIC methods applied on nonfiltered, PCA-based, IG-based, and proposed filtered data, respectively.

to the PCA filter that has the extreme opposite values in most parameters (the lowest value of EPI_{φ}, and the highest value of σ_{Am} and σ_{φ}).

2) Robustness of the Filtering Approach to TomoSAR: To show the influence of spatio-temporal tomographic filtering in the reconstruction of TomoSAR profiles, we first compared the PS point cloud maps and the height maps displayed, respectively, in Figs. 5 and 6, before and after applying the filters. Concerning the vertical localization of the scattering mechanisms in a resolution cell, we found that visually the gradation of the scattered points elevation values varies between approximately 0 and 155 m. This progression is similar to the ground-truth data. We observed after analysis of the projected PS point clouds in Fig. 5 that the spatio-temporal filtering considerably improves the location of the predominant scatterers of certain resolution cells. The points fulfilling the criterion of temporal backscattering stability after being filtered, making it possible to generate maps where the shape of the building is more apparent with a significant points concentration on the two facades and a reduction of points outside the artificial structure whose elevation is poorly estimated.

As for the conventional methods, it is observed that the IG filter has moderately similar performances as to spatio-temporal filters, while the PCA-based filter do not perform very well. The points' concentration increases despite their poor accuracy toward their estimated elevation. Conclusively, the spatio-temporal filters give visually similar good results (note that the height of each PS point was estimated by the standard Maximum Likelihood estimator), but we can say that the MG- and SD-based filters provide more likely better visual aspects compared to PC- and VD-based filters.

On the detection side, the results displayed in Fig. 6 attest to the previous observations. In fact, the number of resolution cells where the H_1 hypothesis is verified decreases when the PCA filter is applied and increases for the remaining filters, particularly in the case of spatio-temporal MG-based, VD-based, and SD-based filters.

To strengthen our analysis, we then proceeded to calculate the percentage of detected scatterers with respect to the total number of pixels, relating to each filtering method. The results presented in Table III show that the highest number of detected PS points belongs to the PCA filter, regardless of its unsatisfactory returns manifested in Fig. 5. Although the detection rate calculated from height maps generated from the GLRT-based detector provides a reliable assessment of the impact the denoising process can have over the tomographic inversion. In view of that measure, the filter based on the SD indicator offers the best PS detectability compared to the other filters, which is consistent with the outcome of Table II discussed previously.

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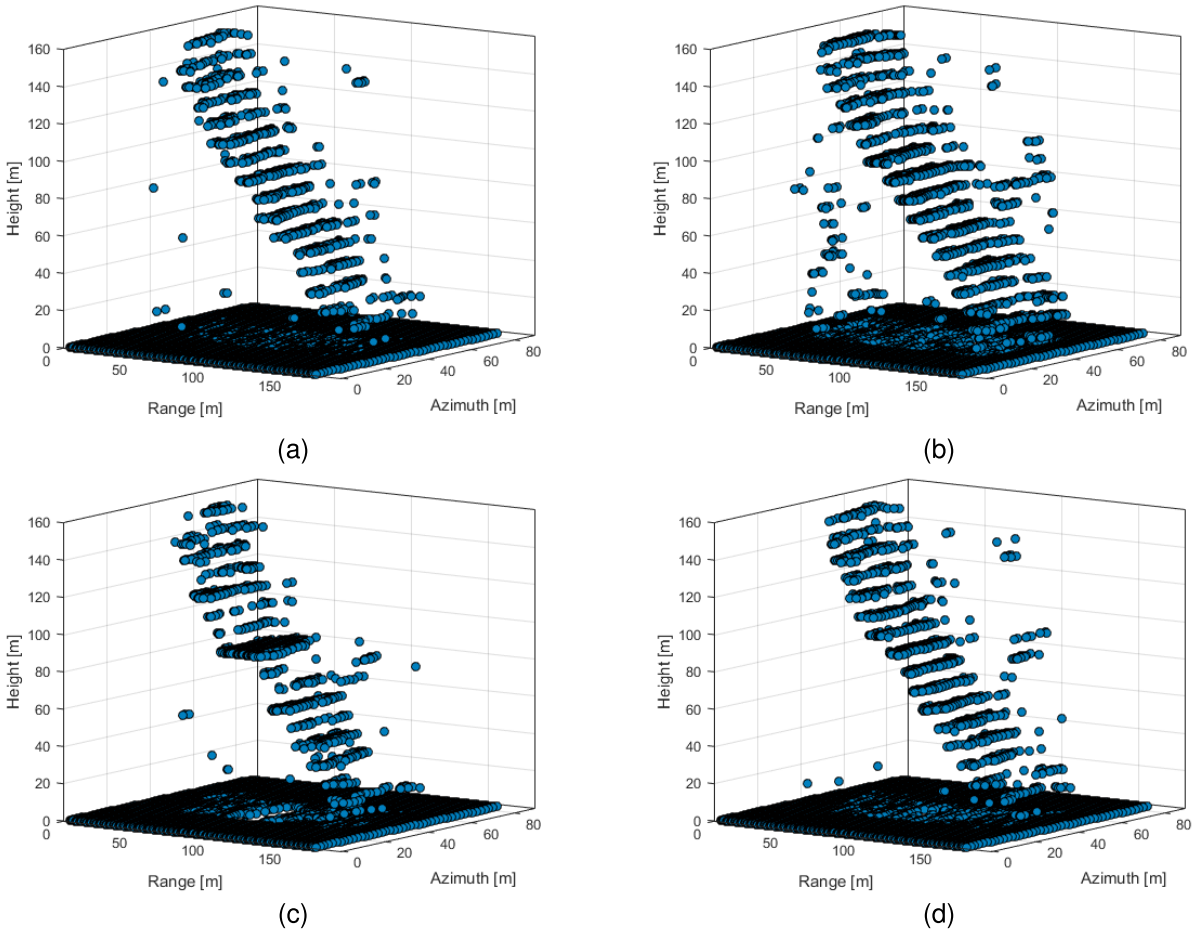


Fig. 9. 3-D point cloud generated using the Robust-CAPON estimator applied on (a) nonfiltered data, (b) PCA-based filtered data, (c) IG-based filtered data, and (d) proposed SD-based filtered data.

It is affirmed that the SD provides the best measure that allows the designed filter to minimize smoothing in outlines and border areas and maximize it in zones with homogeneous structures. As a result, in the remainder of our analysis, we will refer to this spatio-temporal SD-based filter as the proposed denoising method.

3) Effectiveness of the Proposed Spatio-Temporal Filter: Fig. 7 illustrates the tomographic reconstruction results of a considered azimuthal line [see Fig. 7(a)], with the Robust-CAPON and MUSIC methods in the Range-Height plane generated from nonfiltered and filtered data. We can practically see on all tomograms, that the more the value of the range increases the more the pixel elevation decreases, which coincides with the building orientation in the SAR configuration. The choice of the line [colored in green in Fig. 7(a)] relates to the tower structure nature, it covers a part of the facade where the backscattering is relatively weak compared to the other facade. Resolution cells within this profile are considered extremely noisy, so only a few get reconstructed according to the PS criterion, while almost none get detected by GLRT.

First, we investigated the efficiency of the denoising approach toward elevation estimation. The generated tomograms of the Robust-CAPON and MUSIC methods, before any filtering was applied to SAR data, manifest an erroneous estimation of the reflectivity function. Indeed, in the portion where the position in the range is less than 40 m, we notice that the height estimate is incorrect since this section covers a top part of the tower and therefore the elevation values should be beyond 130 m and not around 8–15 m as suggested by the tomograms (b) and (c).

By analyzing the tomograms (d)–(i) resulting from the implementation of the spectral estimators after taking into account the conventional and spatio-temporal filtering approach, we observe that the tomogram generated by the Robust-CAPON method represents a reflectivity whose evolutionary form is linear in the section where the scale in range varies between about 40 up to 160 m, compared to the tomograms generated by MUSIC method which contain several profiles for which the estimate of the elevation is completely erroneous, producing a discontinuity in the height progression of the building. On the other hand, the results indicate that the Robust-CAPON estimator manages to correct the error in the range part less than 40 m after filtering the data and obtains a correct height estimation (the elevation values became around 130 m and up).

Fig. 8 depicts PS height maps of the Mapfre tower generated using both spectral estimators, applied on nonfiltered, PCA-

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based, IG-based, and the proposed SD-based filtered data. The findings reveal that the combination of the proposed denoising method and the Robust-CAPON estimator drives a reliable tomographic reconstruction indicating the robustness of the proposed approach.

To assess these observations further, the completeness C_t and the *R-squared* R_q^2 parameters calculation in Table IV quantify the quality of the height and the reflectivity estimation for the implemented methods. We can deduce that the Robust-CAPON method provides the best results seeing that it has the smallest completeness value and the closest value to 1 of *R-squared*, in both cases (filtered and nonfiltered data). Moreover, both reconstruction methods exhibit better values of C_t and R_q^2 when applied to proposed filtered data compared to the original data.

To evaluate the retained deduction visually, we generated a 3-D point cloud with Robust-CAPON on nonfiltered and filtered data using PCA, IG, and proposed SD-based methods, the results are displayed in Fig. 9. It is observed on one hand that, the PCA filter increases the concentration of points whose elevation is inaccurate. On the other hand, the height progression when the IG filter is applied does not seem sharp enough. We can inspect in Fig. 9(c), the block effect appearing around 70–80 m height. The later demonstrates that the absence of filtering has better outcomes than conventional filters. Simultaneously, the 3-D map in Fig. 9(d) proves that the height estimation is significantly improved by applying the proposed spatio-temporal SD-based method.

V. Conclusion

In this article, we presented a study on the effects of spatio-temporal filtering applied as a preprocessing step for TomoSAR. In this work, an IG parameter was estimated from 3-D PC, MG, VD, and SD quality maps, calculated on blocks structured according to a spatio-temporal rearrangement of pixels. The proposed filtering process was developed and validated using real data. For that, a stack of nine TSX images from ascending orbits covering an urban area of Barcelona was processed. The Mapfre tower was chosen as a case study to analyze the performances of contextual nonlinear filtering on TomoSAR reconstruction and detection.

The experimental results have shown that the considered quality indicators allow labeling each pixel according to the structure it represents. They highlight pixels in areas where maximum filtering should be applied and do the opposite in transitioning areas to preserve edges by applying minimum smoothing. Performance analysis concluded that the designed filters respect the required characteristics for reliable amplitude and phase information preservation. In particular, the SD-based filter generates slightly better results compared to the other three indicators and the classical IG and PCA filters. Additionally, the produced tomograms of the selected azimuthal profile along with the generated height maps demonstrated the ability of the proposed filter combined with the Robust-CAPON estimator to provide an accurate height estimation and a reduced impact of noise in the 2-D and 3-D point clouds.

Future works could consider addressing the effect of spatio-temporal filtering, in multiple and volumetric scatterers detection, to form 3-D height maps in urban and vegetated areas, respectively. An extension of the proposed approach to 4-D TomoSAR can also be investigated.

Appendix

The second differences az, rg, nr, and D, in (12), can be calculated for each pixel value P, according to the following formulas [50]:

$$\mathrm{az}^n(x,r) = P(x-1,r,n)+P(x+1,r,n)-2\cdot P(x,r,n) \quad (21)$$

$$\mathrm{rg}^n(x,r) = P(x,r-1,n)+P(x,r+1,n)-2\cdot P(x,r,n) \quad (22)$$

$$\mathrm{nr}^n(x,r) = P(x,r,n-1)+P(x,r,n+1)-2\cdot P(x,r,n) \quad (23)$$

$$D_1^n(x,r) = P(x-1,r-1,n)+P(x+1,r+1,n)-2\cdot P(x,r,n)$$

$$D_2^n(x,r) = P(x+1,r-1,n)+P(x-1,r+1,n)-2\cdot P(x,r,n)$$

$$D_3^n(x,r) = P(x-1,r-1,n-1)+P(x+1,r+1,n+1) -2\cdot P(x,r,n)$$

$$D_4^n(x,r) = P(x,r-1,n-1)+P(x,r+1,n+1) -2\cdot P(x,r,n)$$

$$D_5^n(x,r) = P(x+1,r-1,n-1)+P(x-1,r+1,n+1) -2\cdot P(x,r,n)$$

$$D_6^n(x,r) = P(x-1,r,n-1)+P(x+1,r,n+1)-2\cdot P(x,r,n)$$

$$D_7^n(x,r) = P(x-1,r+1,n-1)+P(x+1,r-1,n+1) -2\cdot P(x,r,n)$$

$$D_8^n(x,r) = P(x+1,r,n-1)+P(x-1,r,n+1)-2\cdot P(x,r,n)$$

$$D_9^n(x,r) = P(x,r+1,n-1)+P(x,r-1,n+1)-2\cdot P(x,r,n)$$

$$D_{10}^n(x,r) = P(x+1,r+1,n-1) +P(x-1,r-1,n+1)-2\cdot P(x,r,n). \quad (24)$$

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